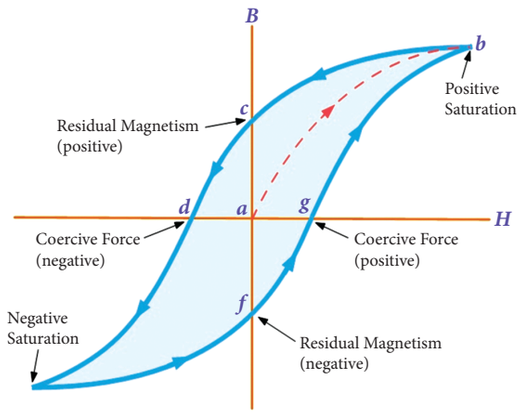


Fig 3.17 *Hysteresis Loop*

When H is reversed by reversing the current through the solenoid, then B is reduced to zero at point C where H = OC. This value is required to clear off the residual magnetism. This is known as the coercive force and is a measure of the coercivity of the material.

After reducing the magnetism to zero, if the value of H is further increased in the negative direction (i.e reversed direction), the iron bar reaches a state of magnetic saturation at point A_1, which is negative saturation (H = OL). By taking H back from its value corresponding to negative saturation (OL) to its value for positive saturation (OM), the closed loop which is obtained when iron bar is taken through one complete cycle of magnetism. This loop is called Hysteresis Loop .

In this BH curve, it is seen that B always lag behind H. The two never attain zero value simultaneously .Hysteresis literally means to lag behind . The closed loop OAR_1, $CA_1R_2C_1A$ which is obtained when iron bar is taken through one complete cycle of reversal of magnetisation is known as Hysteresis loop.

Hysteresis Loss

- It is the loss of power due to hysteresis and expressed in watts or KW.
- Hysteresis cannot be avoided but can be minimised by selecting proper metal.
- Lesser the hysteresis constant, better the metal for A.C electromagnet.
- So, usually silicon steel is used for A.C circuit as it's hysteresis constant is 0.001.

3.7 RULES/LAWS RELATED TO MAGNETISM

Rule or Law	Uses
Cork Screw Rule	To find out the direction of line of force (magnetic field) around a straight current carrying conductor.
Helix Rule	To find out polarity of the poles of an electromagnet (solenoid)
End Rule	To find out polarity of the poles of an electromagnet (solenoid)
Fleming's Right Hand Rule	To find out the direction of current in the conductor of a generator.
Fleming's Left Hand Rule	To find out the direction of rotation of the armature of D.C motor.
Lenz's Law	To find out the direction of the counter current produced in the armature.
Ampere rule	To find out the direction of line of force around the current carrying conductor.

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